Additive Bases from Primitive Dyck Words: Regular Underapproximations, Motzkin Coding, and Digit Lifting

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Abstract

We study additive representations by integers whose canonical binary expansions are primitive Dyck words. Pairing consecutive bits yields a positional form of the classical relation between Dyck paths and two-coloured Motzkin paths: except for 10, primitive Dyck words are exactly the binary block images of base-4 words $3w0$, where w is a two-coloured Motzkin word. This exposes a regular underapproximation, digit closure, and sharp generation bounds. We prove an interval digit-lifting theorem for digitally closed sets and a constructive base-4 propagation algorithm that lifts finite sumset certificates to infinite tails in logarithmically many recursive stages. Combining these tools with exact finite certificates and generation-gap lower bounds, we classify all positive even integers requiring more than six primitive Dyck summands. The integer 46 requires eight, and $34, 44, 98, 154, 198, 202, 206, 838, 842, 846$ require seven; every other positive even integer requires at most six. Thus 848 is the sharp eventual threshold. The bound is asymptotically optimal because $10 \cdot 4^{k+1} - 6$ requires six summands for every $k \geq 2$. The associated halved family has exact asymptotic additive order five. Supplementary programs reproduce all finite certificates using exact integer arithmetic.

Keywords: Dyck language; primitive Dyck word; deterministic context-free language; regular underapproximation; additive basis; two-coloured Motzkin word; numeration system; digit lifting.
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1 Introduction

Numeration systems provide a natural interface between formal-language theory and additive number theory. Given a language L over a digit alphabet, one may read the words of L as integers and ask whether the resulting value set is an additive basis, what its least order is, and whether the exceptional targets can be classified effectively. For regular languages, these questions can often be approached through finite automata. Bell, Hare, and Shallit gave general results for automatic additive bases [1], and related methods have produced sharp results for binary palindromes [10] and binary squares [5].

Bell, Lidbetter, and Shallit applied regular-language approximation to the language of balanced binary words and obtained strong additive theorems for the corresponding totally balanced numbers [2]. Their work is a particularly close predecessor of the present paper. Regular approximations of context-free languages have a broader history, including constructions of both underapproximations and overapproximations by finite automata [8, 6].

Language-based numeration has also been studied in the framework of abstract numeration systems [9]. In particular, Charlier, Le Gonidec, and Rigo considered a generalized numeration system built on the Dyck language [3]. Their numerical map is induced by an ordering of the underlying language, whereas the present paper reads each word by its ordinary positional binary value. On the combinatorial side, bijections relating Dyck paths or ordered trees to two-coloured Motzkin paths are classical [4]. Our use of Motzkin paths is tailored to the paired binary digits and hence retains exact base-2/base-4 positional values.

The proof below is hybrid in a precise but deliberately limited sense. An explicit regular underapproximation supplies the six-summand covering used for targets divisible by 4. For targets congruent to 2 modulo 4, our argument instead uses a five-summand covering by the full Motzkin-coded family. We do not claim that every possible regular underapproximation must fail for the latter residue class.

Let $\mathcal{D}$ be the Dyck language over $\{1,0\}$, with 1 interpreted as an up-step and 0 as a down-step. A nonempty Dyck word is *primitive* if its path returns to height zero only at the endpoint. Let $\mathcal{D}_{\mathrm{P}}$ denote the language of primitive Dyck words. Since $\mathcal{D}_{\mathrm{P}} = 1\mathcal{D}0$, it is deterministic context-free. It is not regular, because

$$\mathcal{D}_{\mathrm{P}} \cap 1^*0^* = \{1^n0^n : n \geq 1\}.$$

We read the words of $\mathcal{D}_{\mathrm{P}}$ as binary integers and adjoin 0; the resulting set is denoted by $\mathcal{N}_{\mathrm{pD}}$.

The primitive restriction is substantial. Every nonempty Dyck word factors uniquely into primitive factors, but an additive representation in our problem requires each summand itself to be represented by a single primitive word. Thus a representation by unrestricted balanced words does not preserve the number of summands after passage to primitive factors. Moreover, the language $\mathcal{D}_{\mathrm{P}}$ is not available to finite-automaton methods in its entirety.

The key structural step is a positional block coding. Pairing consecutive binary digits produces base-4 digits, and the primitive condition becomes a Motzkin prefix condition. We prove the exact language identity

$$\mathcal{D}_{\mathrm{P}} = \{10\} \cup \phi(3\mathsf{Motz}0),$$

where $\phi(0) = 00$, $\phi(1) = 01$, $\phi(2) = 10$, and $\phi(3) = 11$. The novelty used here is not the general existence of Dyck–2-Motzkin bijections, but the compatibility of this primitive-word block map with canonical positional value. The regular language $11(01 \mid 10)^*00$ then supplies one additive covering, while the full Motzkin-coded family supplies the digit closure used in the other covering.

Our contributions are fourfold.

(i) We adapt the classical Dyck–2-Motzkin viewpoint to canonical binary valuation and obtain an exact base-4 characterization of the primitive Dyck value set, together with sharp generation bounds.

(ii) We separate two complementary roles in the additive proof: an explicit regular underapproximation covers the multiples of 4, while a five-summand theorem for the full Motzkin-coded family covers the auxiliary targets needed in the residue class 2 modulo 4.

(iii) We formulate a general interval digit-lifting theorem, prove a base-4 tail-propagation principle, and extract a constructive algorithm that converts a finite table of seed representations into representations throughout the infinite tail.

(iv) We determine the complete exceptional set, the sharp eventual threshold, and the exact relative asymptotic order; generation gaps also give an infinite lower-bound sequence.

For a positive even integer N, let $r(N)$ be the least number of positive primitive Dyck numbers whose sum is N. This is well defined because $10 \in \mathcal{D}_{\mathrm{P}}$ represents 2, so every positive even N is a sum of $N/2$ copies of 2. Our first main result classifies exactly the values above level six.

Theorem 1. *The values of $r(N)$ exceeding six are exactly as follows:*

$$r(46) = 8,$$
$$r(N) = 7 \iff N \in \{34, 44, 98, 154, 198, 202, 206, 838, 842, 846\}.$$

Every other positive even integer satisfies $r(N) \le 6$. Equivalently, 848 is the least even integer T such that every even $N \ge T$ is a sum of at most six primitive Dyck numbers.

The eventual bound cannot be lowered.

Theorem 2. *Let*

$$h_{\mathrm{asym}} = \min\{h : \textit{every sufficiently large even } N \textit{ satisfies } r(N) \le h\}.$$

Then $h_{\mathrm{asym}} = 6$. More precisely, for every integer $k \ge 2$,

$$N_k = 10 \cdot 4^{k+1} - 6$$

satisfies $r(N_k) = 6$.

Section 2 fixes the language and additive notation. Section 3 proves the positional Motzkin block coding and generation bounds. Section 4 develops digit lifting, its constructive tail algorithm, the regular underapproximation, and the finite certificates. Section 5 proves the complete classification, and Section 6 proves asymptotic optimality. The supplementary verification algorithms are described in Section 7.

2 Languages, values, and additive notation

A word $w \in \{1, 0\}^*$ is a *Dyck word* if it contains equally many 1's and 0's and every prefix contains at least as many 1's as 0's. The empty word is denoted by λ. A nonempty Dyck word is *primitive* if no proper nonempty prefix is balanced. Equivalently, its height path first returns to zero at the endpoint. We write $\mathcal{D}$ for the Dyck language and $\mathcal{D}_{\mathrm{P}}$ for its primitive sublanguage. Every primitive word is uniquely of the form $1u0$ with $u \in \mathcal{D}$.

For a word w over the base-b digits, let $[w]_b$ denote its numerical value. For a language L over those digits, write

$$\mathrm{val}_b(L) = \{[w]_b : w \in L\}.$$

Define

$$\mathcal{N}_{\mathrm{pD}} = \{0\} \cup \{[w]_2 : w \in \mathcal{D}_{\mathrm{P}}\} \subseteq \mathbb{Z}_{\ge 0}.$$

Every positive member of $\mathcal{N}_{\mathrm{pD}}$ is even. The following stronger terminal property will be used repeatedly.

Lemma 3. *The number 2 is the only positive primitive Dyck number congruent to 2 modulo 4. Every other positive primitive Dyck number is divisible by 4.*

Proof. Let $w \in \mathcal{D}_{\mathrm{P}}$. Its last letter is 0. If $|w| = 2$, then $w = 10$ and $[w]_2 = 2$. Suppose $|w| \ge 4$ and write $w = ub0$ with $b \in \{0, 1\}$. If $b = 1$, the suffix 10 has net height zero, so the proper prefix u is balanced, contradicting primitivity. Hence $b = 0$, the word ends in 00, and $4 \mid [w]_2$. □

Among the C_n Dyck words of length $2n$, exactly C_{n-1} are primitive, so their proportion tends to $1/4$. Thus primitive words are not sparse at a fixed length; the additive obstructions below arise from their hierarchical value structure rather than from a vanishing language density.

For a nonnegative integer k and a set X of nonnegative integers, let kX denote the set of sums of exactly k not necessarily distinct members of X, with $0X = \{0\}$. Put

$$F_k(X) = \bigcup_{j=0}^{k} jX.$$

Thus $F_k(X)$ is the set of sums of at most k elements of X. If $0 \in X$, membership in kX may be read as a representation by at most k positive elements of X: the zero entries are only padding and are deleted when the representation is transferred back to positive primitive Dyck summands. This is not dilation notation: for a scalar c, we write $c \cdot X = \{cx : x \in X\}$.

Lemma 3 motivates the halved positive family

$$\mathcal{P} = \{v/4 : v \in \mathcal{N}_{\mathrm{pD}},\ v \geq 12\}. \tag{1}$$

Then

$$\mathcal{N}_{\mathrm{pD}} = \{0, 2\} \cup 4 \cdot \mathcal{P}. \tag{2}$$

The first values in $\mathcal{P}$ are

$$3, 13, 14, 53, 54, 57, 58, 60, 213, 214, 217, \ldots.$$

3 Two-coloured Motzkin block coding

Two-coloured Motzkin paths are classically related bijectively to Dyck paths and ordered trees; see, for example, Deutsch and Shapiro [4]. The construction below is a direct left-to-right block map for primitive Dyck words. Its role in this paper is value-sensitive: the same pairs are simultaneously the base-4 digits of the represented integer.

Binary words of even length may be read blockwise in base 4 through

$$00 \leftrightarrow 0, \qquad 01 \leftrightarrow 1, \qquad 10 \leftrightarrow 2, \qquad 11 \leftrightarrow 3.$$

Define the uniform morphism

$$\phi : \{0, 1, 2, 3\}^* \longrightarrow \{0, 1\}^*$$

by

$$\phi(0) = 00, \quad \phi(1) = 01, \quad \phi(2) = 10, \quad \phi(3) = 11.$$

Assign weights

$$\sigma(3) = 1, \qquad \sigma(0) = -1, \qquad \sigma(1) = \sigma(2) = 0,$$

and extend σ additively to words. A word over $\{0, 1, 2, 3\}$ is a *two-coloured Motzkin word* if every prefix has nonnegative weight and its total weight is zero. Let Motz denote this language; it contains λ.

Theorem 4 (Block-coding characterization)**.** *The language of primitive Dyck words satisfies the disjoint identity*

$$\mathcal{D}_{\mathrm{P}} = \{10\} \cup \phi\big(3\mathsf{Motz}0\big). \tag{3}$$

Proof. The word 10 is primitive. Let $B \in \mathcal{D}_{\mathrm{P}}$ have length at least 4, and group its bits into pairs. Write

$$B = \phi(q_1 q_2 \cdots q_L), \qquad q_i \in \{0, 1, 2, 3\}.$$

Let $h_B(t)$ be the binary height after t bits. Across a complete block,

$$h_B(2j) - h_B(2j-2) = 2\sigma(q_j). \tag{4}$$

Primitivity forces the first block to be 11 and the last block to be 00; otherwise the path would return to height zero at position 2 or at position $2L-2$. Hence $q_1 = 3$ and $q_L = 0$.

Put $u = q_2 \cdots q_{L-1}$. At each internal block boundary, $h_B(2j)$ is a positive even integer. Summing (4) therefore gives

$$\sum_{i=1}^{j} \sigma(q_i) \geq 1 \qquad (1 \leq j \leq L-1).$$

Subtracting $\sigma(q_1) = 1$ shows that every prefix of u has nonnegative weight. Since the total block weight is zero and the endpoint weights are 1 and -1, the total weight of u is zero. Thus $u \in \mathsf{Motz}$ and $B = \phi(3u0)$.

Conversely, let $u = q_2 \cdots q_m \in \mathsf{Motz}$ and set $B = \phi(3u0)$. At each nonterminal even block boundary the binary height is

$$2\left(1 + \sum_{i=2}^{j} \sigma(q_i)\right) \geq 2,$$

and the total height is zero. It remains to inspect odd positions. A block 3 or 2 begins with 1 and therefore increases the height. A block 1 begins with 0, but it starts at an even boundary of height at least 2, so the intermediate height is at least 1. If a block 0 occurs inside u, then the Motzkin weight after reading that block must remain nonnegative. Since the block has weight -1, its preceding weight is therefore at least 1; hence the block starts at height at least 4 and its first 0 leaves positive height. Finally, the appended terminal block 0 starts at height 2, passes through height 1, and ends at 0. Thus every proper prefix of B has positive height, so $B \in \mathcal{D}_{\mathrm{P}}$. □

For a positive integer n, let $\mathrm{quad}(n)$ denote its standard base-4 expansion.

Corollary 5 (Base-4 value characterization)**.** *A positive integer p lies in $\mathcal{P}$ if and only if there exists $w \in \mathsf{Motz}$ such that*

$$\mathrm{quad}(p) = 3w. \tag{5}$$

Proof. By definition, $p \in \mathcal{P}$ exactly when $4p$ is represented by a primitive Dyck word of length at least 4. Multiplication by 4 appends the base-4 digit 0. Theorem 4 therefore says exactly that the base-4 expansion of $4p$ is $3w0$ with $w \in \mathsf{Motz}$, equivalently that $\mathrm{quad}(p) = 3w$. □

Corollary 6. *If $p \in \mathcal{P}$, then $4p+1$ and $4p+2$ also belong to $\mathcal{P}$.*

Proof. By Corollary 5, there exists $w \in \mathsf{Motz}$ such that

$$\mathrm{quad}(p) = 3w.$$

Then

$$\mathrm{quad}(4p+1) = 3w1, \qquad \mathrm{quad}(4p+2) = 3w2.$$

Since

$$\sigma(1) = \sigma(2) = 0,$$

appending either digit preserves membership in Motz. The claim follows from Corollary 5. □

We next determine the smallest and largest values in each generation.

Lemma 7 (Extremal two-coloured Motzkin values)**.** *For every $\ell \geq 0$ and every two-coloured Motzkin word w of length ℓ,*

$$\frac{4^\ell - 1}{3} \leq [w]_4 \leq 4^\ell - 2^\ell. \tag{6}$$

Both bounds are attained. The lower bound is attained by 1^ℓ, and the upper bound is attained by

$$w_{\max,\ell} = \begin{cases} 3^a 0^a, & \ell = 2a, \\ 3^a 2\, 0^a, & \ell = 2a+1. \end{cases}$$

Here 1^0 is understood to be the empty word λ.

Proof. We use induction on ℓ. The assertion is immediate for $\ell = 0$. Write

$$w = c_0 c_1 \cdots c_{\ell-1}, \qquad c_j \in \{0,1,2,3\}.$$

Lower bound. Since $\sigma(0) = -1$, the first digit cannot be 0. If $c_0 \in \{1,2\}$, then $\sigma(c_0) = 0$, so deleting the first digit does not change any subsequent prefix weight or the total weight. Hence the suffix $w' = c_1 \cdots c_{\ell-1}$ is again a two-coloured Motzkin word. By the induction hypothesis for words of length $\ell - 1$,

$$[w]_4 \geq 4^{\ell-1} + [w']_4 \geq 4^{\ell-1} + \frac{4^{\ell-1} - 1}{3} = \frac{4^\ell - 1}{3}.$$

If $c_0 = 3$, then

$$[w]_4 \geq [3\,\underbrace{00\cdots0}_{\ell-1 \text{ digits}}]_4 = 3 \cdot 4^{\ell-1} \geq \frac{4^\ell - 1}{3}.$$

Equality is attained by $w = 1^\ell$.

Upper bound. If $c_0 \in \{1,2\}$, then, as above, w' is again a two-coloured Motzkin word. By the induction hypothesis for words of length $\ell - 1$,

$$\begin{aligned} [w]_4 &= [c_0 c_1 \cdots c_{\ell-1}]_4 \\ &\leq [2c_1 \cdots c_{\ell-1}]_4 \\ &\leq 2 \cdot 4^{\ell-1} + \left(4^{\ell-1} - 2^{\ell-1}\right) \\ &= 3 \cdot 4^{\ell-1} - 2^{\ell-1} \\ &\leq 4^\ell - 2^\ell. \end{aligned}$$

Now suppose $c_0 = 3$, and let r be the length of the shortest nonempty prefix of w having total weight 0. This prefix has the form $3u0$, where

$$u = c_1 \cdots c_{r-2},$$

and let

$$v = c_r \cdots c_{\ell-1}$$

be the remaining suffix. Then

$$2 \leq r \leq \ell, \qquad |u| = r - 2, \qquad |v| = \ell - r,$$

and $u, v \in \mathsf{Motz}$. Thus $w = 3u0v$. By place value and the induction hypothesis,

$$\begin{aligned}
[w]_4 &= 3 \cdot 4^{\ell-1} + [u]_4\, 4^{\ell-r+1} + [v]_4 \\
&\leq 3 \cdot 4^{\ell-1} + \big(4^{r-2} - 2^{r-2}\big)4^{\ell-r+1} + 4^{\ell-r} - 2^{\ell-r} \\
&= 4^{\ell} - 2^{2\ell-r} + 2^{2\ell-2r} - 2^{\ell-r} \\
&= 4^{\ell} - \big(2^{2\ell-r} - 2^{2\ell-2r}\big) - 2^{\ell-r}.
\end{aligned}$$

It remains to verify

$$2^{2\ell-r} - 2^{2\ell-2r} \geq 2^{\ell} - 2^{\ell-r}.$$

After factoring, this is

$$2^{2\ell-2r}(2^r - 1) \geq 2^{\ell-r}(2^r - 1),$$

which follows from $r \leq \ell$. This proves the upper bound.

Finally, the displayed words $w_{\max,\ell}$ are two-coloured Motzkin words. A geometric-series calculation gives

$$[w_{\max,\ell}]_4 = 4^{\ell} - 2^{\ell},$$

so the upper bound is attained. □

An element of $\mathcal{P}$ belongs to *generation* d if its base-4 expansion has exactly d digits. By Corollary 5, a generation-d element p satisfies

$$\mathrm{quad}(p) = 3w$$

for some $w \in \mathsf{Motz}$ with $|w| = d-1$. Therefore

$$p = [3w]_4 = 3 \cdot 4^{d-1} + [w]_4. \tag{7}$$

The following corollary is now immediate from Lemma 7.

Corollary 8 (Generation bounds)**.** *For every $d \geq 1$, every generation-d element of $\mathcal{P}$ lies in*

$$[g_d, U_d] = \left[3 \cdot 4^{d-1} + \frac{4^{d-1}-1}{3}, 4^d - 2^{d-1}\right], \tag{8}$$

and both endpoints are attained. In particular,

(a) there is no element of $\mathcal{P}$ in $(U_d, 3 \cdot 4^d)$;

(b) for every $k \geq 0$,

$$3g_{k+1} = 10 \cdot 4^k - 1. \tag{9}$$

Proof. Let p be a generation-d element. Applying (6) with $\ell = d-1$ gives

$$\frac{4^{d-1}-1}{3} \leq [w]_4 \leq 4^{d-1} - 2^{d-1}.$$

Substituting these bounds into (7) yields (8). The lower endpoint is attained by $w = 1^{d-1}$, and the upper endpoint is attained by $w = w_{\max,d-1}$. Part (a) follows because the next generation begins at $g_{d+1} > 3 \cdot 4^d$. Finally,

$$3g_{k+1} = 3\left(3 \cdot 4^k + \frac{4^k-1}{3}\right) = 10 \cdot 4^k - 1.$$

□

4 Regular underapproximation, digit lifting, and finite certificates

We first isolate a digit-lifting mechanism that is independent of Dyck words.

Theorem 9 (Interval digit lifting)**.** *Let $b \geq 2$, let $\alpha \leq \beta$ be integers, and let $A \subseteq \mathbb{Z}$ satisfy*

$$b \cdot A + [\alpha, \beta] \subseteq A. \tag{10}$$

For every $k \geq 1$ and every $m \in kA$,

$$[bm + k\alpha, bm + k\beta] \subseteq kA. \tag{11}$$

Proof. Write $m = a_1 + \cdots + a_k$ with $a_i \in A$. Every integer $r \in [k\alpha, k\beta]$ is a sum $r = \eta_1 + \cdots + \eta_k$ with $\eta_i \in [\alpha, \beta]$: start with all $\eta_i = \alpha$ and distribute $r - k\alpha$ unit increments among the k coordinates, never exceeding β. By (10), $ba_i + \eta_i \in A$, and hence

$$bm + r = \sum_{i=1}^{k} (ba_i + \eta_i) \in kA.$$

$\square$

Corollary 10 (Base-4 digit lifting)**.** *Let $k \geq 1$, and let $A \subseteq \mathbb{Z}_{>0}$ satisfy*

$$4 \cdot A + \{1, 2\} \subseteq A. \tag{12}$$

If $m \in kA$, then

$$[4m + k, 4m + 2k] \subseteq kA. \tag{13}$$

Proof. Apply Theorem 9 with $b = 4$, $\alpha = 1$, and $\beta = 2$. $\square$

Corollary 11 (Propagation from a finite interval to a tail)**.** *Let $k \geq 3$, and let $A \subseteq \mathbb{Z}_{>0}$ satisfy* (12)*. Suppose that, for some integer $M \geq 1$,*

$$[M, 4M + k - 1] \subseteq kA. \tag{14}$$

Then

$$[M, \infty) \subseteq kA.$$

Proof. We use strong induction on $n \geq M$. The seed interval handles $M \leq n \leq 4M + k - 1$. Let $n \geq 4M + k$, and choose the unique

$$\rho \in \{k, k+1, k+2, k+3\}$$

with $\rho \equiv n \pmod 4$. Since $k \geq 3$, $k \leq \rho \leq k + 3 \leq 2k$. Put $m = (n - \rho)/4$. Then m is an integer and

$$m \geq \frac{4M + k - (k+3)}{4} = M - \frac{3}{4},$$

so $m \geq M$. Also $m < n$. The induction hypothesis gives $m \in kA$, and Corollary 10 gives

$$n = 4m + \rho \in [4m + k, 4m + 2k] \subseteq kA.$$

$\square$

Corollary 12 (Constructive tail representations). *Under the hypotheses of Corollary 11, suppose that one stores a representation*

$$n = a_1 + \cdots + a_k, \qquad a_i \in A,$$

for every $n \in [M, 4M + k - 1]$. Then a k-term representation of every $n \geq M$ can be constructed recursively. The construction uses $O(\log(n/M))$ recursive stages and $O(k \log(n/M))$ applications of the digit maps $a \mapsto 4a + 1$ and $a \mapsto 4a + 2$.

Proof. For n in the seed interval, return the stored representation. Otherwise choose ρ and $m = (n - \rho)/4$ as in the proof of Corollary 11, and recursively write $m = a_1 + \cdots + a_k$. Since $k \leq \rho \leq 2k$, choose digits $\eta_i \in \{1, 2\}$ whose sum is ρ; explicitly, take $\rho - k$ of the digits equal to 2 and the remaining digits equal to 1. Then

$$n = \sum_{i=1}^{k} (4a_i + \eta_i),$$

and every transformed summand belongs to A by (12). At each nonseed stage the recursive argument is strictly less than one quarter of the current target. Hence the recursion reaches the seed interval after at most $1 + \lceil \log_4(n/M) \rceil$ stages, and each stage performs k digit updates. $\square$

4.1 A regular underapproximation for multiples of four

Consider the regular language

$$\mathcal{R} = 11(01 \mid 10)^*00.$$

By Theorem 4, one has $\mathcal{R} \subseteq \mathcal{D}_{\mathrm{P}}$: after block decoding, its words are exactly $\phi(3\{1, 2\}^*0)$, and every word over $\{1, 2\}$ is a two-coloured Motzkin word. Thus $\mathcal{R}$ is an explicit regular underapproximation of the nonregular language $\mathcal{D}_{\mathrm{P}}$.

Define the corresponding regular base-4 value family

$$\mathcal{E} = \{0\} \cup \big\{ [3\varepsilon_1 \cdots \varepsilon_j]_4 : j \geq 0,\ \varepsilon_i \in \{1, 2\} \big\},$$

and put $\mathcal{E}^+ = \mathcal{E} \setminus \{0\}$. When $j = 0$, the suffix is the empty word, and the corresponding element is $[3]_4 = 3$. Every word $\varepsilon_1 \cdots \varepsilon_j$ belongs to Motz. Hence Corollary 5 gives

$$\mathcal{E}^+ \subseteq \mathcal{P}.$$

More precisely, $\mathrm{val}_2(\mathcal{R}) = 4 \cdot \mathcal{E}^+$. It follows from (2) that

$$4 \cdot \mathcal{E} \subseteq \mathcal{N}_{\mathrm{pD}}. \tag{15}$$

If $e \in \mathcal{E}^+$, then its base-4 expansion has the form

$$e = [3\varepsilon_1 \cdots \varepsilon_j]_4, \qquad \varepsilon_i \in \{1, 2\}.$$

Consequently,

$$4e + 1 = [3\varepsilon_1 \cdots \varepsilon_j 1]_4, \qquad 4e + 2 = [3\varepsilon_1 \cdots \varepsilon_j 2]_4.$$

Both integers again belong to $\mathcal{E}^+$. Therefore

$$4 \cdot \mathcal{E}^+ + \{1, 2\} \subseteq \mathcal{E}^+. \tag{16}$$

We use only the following finite inclusions as computational certificates.

Lemma 13 (Finite certificate for the regular family)**.** *Put*

$$\begin{aligned} A_0 &= \{0, 3, 13, 14\}, & B &= \{53, 54, 57, 58\}, \\ A &= \{3, 13, 14, 53, 54, 57, 58\}, & C &= \{213, 214, 217, 218, 229, 230, 233, 234\}. \end{aligned}$$

Then $A_0 \subseteq \mathcal{E}$ and $A, B, C \subseteq \mathcal{E}^+$. Moreover,

sumset	*contained interval*
$6A_0$	$[25, 62]$
$B + 5A_0$	$[56, 128]$
$2B + 4A_0$	$[106, 172]$
$3B + 3A_0$	$[159, 216]$
$4B + 2A_0$	$[212, 260]$

(17)

and

sumset	*contained interval*
$6A$	$[218, 260]$
$C + 5A$	$[258, 524]$
$2C + 4A$	$[446, 700]$
$3C + 3A$	$[648, 876]$
$4C + 2A$	$[858, 1052]$

(18)

hold.

Proof. Each assertion is an exhaustive finite computation of a sumset. For a finite set X, start with $S_0(X) = \{0\}$ and compute recursively

$$S_{i+1}(X) = S_i(X) + X.$$

The supplementary verification script checks that every integer in each displayed interval belongs to the corresponding sumset. □

Proposition 14. *Every integer $n \geq 25$ belongs to $6\mathcal{E}$.*

Proof. The five intervals in (17) overlap and give

$$[25, 217] \subseteq 6\mathcal{E}.$$

Every sumset occurring in (18) is formed only from the sets $A, C \subseteq \mathcal{E}^+$; no zero summand is used. Hence the five intervals in that certificate overlap to give

$$[218, 877] \subseteq 6\mathcal{E}^+.$$

Combining the two finite ranges, we obtain

$$[25, 877] \subseteq 6\mathcal{E}. \tag{19}$$

We apply Corollary 11 to $A = \mathcal{E}^+$, rather than to $A = \mathcal{E}$. The set $\mathcal{E}$ contains 0 and therefore does not satisfy (12), since that condition would force $1, 2 \in \mathcal{E}$. By contrast, $\mathcal{E}^+$ satisfies (12) by (16). Since

$$877 = 4 \cdot 218 + 6 - 1,$$

the inclusion

$$[218, 877] \subseteq 6\mathcal{E}^+$$

is exactly the seed condition with $k = 6$ and $M = 218$. Corollary 11 therefore gives

$$[218, \infty) \subseteq 6\mathcal{E}^+ \subseteq 6\mathcal{E}.$$

Combining this with

$$[25, 217] \subseteq 6\mathcal{E}$$

proves

$$[25, \infty) \subseteq 6\mathcal{E}.$$

□

It follows from Proposition 14 and (15) that every multiple of 4 at least 100 is a sum of at most six primitive Dyck numbers.

4.2 Five-summand covering by the full context-free family

The regular family $\mathcal{E}$ supplies the covering used for integers congruent to 0 modulo 4. For integers congruent to 2 modulo 4, our proof instead uses five-summand representations from the full halved primitive family $\mathcal{P}$. This establishes the complementary role of the full Motzkin-coded family in the present argument; it is not an impossibility statement about all conceivable regular subfamilies.

We now verify explicitly the small elements of $\mathcal{P}$ used in the finite interval certificate. For

$$w = c_1 c_2 \cdots c_\ell \in \{0, 1, 2, 3\}^\ell,$$

let

$$\mathbf{h}(w) := \big(\sigma(c_1),\, \sigma(c_1c_2),\, \ldots,\, \sigma(c_1c_2\cdots c_\ell)\big)$$

be the sequence of weights of its nonempty prefixes. Recall that

$$\sigma(3) = 1, \qquad \sigma(0) = -1, \qquad \sigma(1) = \sigma(2) = 0.$$

For the empty word, put $\mathbf{h}(\lambda) = ()$. In this notation, $w \in \mathsf{Motz}$ if and only if every component of $\mathbf{h}(w)$ is nonnegative and its last component is 0.

First, we enumerate all two-coloured Motzkin words of lengths 0, 1, and 2. The corresponding elements of $\mathcal{P}$ are as follows:

$\lvert w\rvert$	w	$[3w]_4$	$\mathbf{h}(w)$
0	λ	$[3]_4 = 3$	$()$
1	1	$[31]_4 = 13$	(0)
1	2	$[32]_4 = 14$	(0)
2	11	$[311]_4 = 53$	$(0, 0)$
2	12	$[312]_4 = 54$	$(0, 0)$
2	21	$[321]_4 = 57$	$(0, 0)$
2	22	$[322]_4 = 58$	$(0, 0)$
2	30	$[330]_4 = 60$	$(1, 0)$

This table is complete. A word of length 1 has total weight 0 only when its unique letter is 1 or 2. For a word of length 2, if the first letter is 1 or 2, then the second letter must again be 1 or 2, giving

$$11,\ 12,\ 21,\ 22.$$

If the first letter is 3, then the second letter must be 0 in order to return from height 1 to total weight 0, giving the word 30. A word beginning with 0 is impossible because its first prefix has negative weight. Thus there are exactly

$$1 + 2 + 5 = 8$$

two-coloured Motzkin words of lengths 0, 1, and 2.

By Corollary 5, the corresponding elements of $\mathcal{P}$ are exactly

$$L = \{3, 13, 14, 53, 54, 57, 58, 60\} \subseteq \mathcal{P}. \tag{20}$$

Next, we enumerate all two-coloured Motzkin words of length 3:

w	$[3w]_4$	$\mathbf{h}(w)$
111	$[3111]_4 = 213$	$(0,0,0)$
112	$[3112]_4 = 214$	$(0,0,0)$
121	$[3121]_4 = 217$	$(0,0,0)$
122	$[3122]_4 = 218$	$(0,0,0)$
130	$[3130]_4 = 220$	$(0,1,0)$
211	$[3211]_4 = 229$	$(0,0,0)$
212	$[3212]_4 = 230$	$(0,0,0)$
221	$[3221]_4 = 233$	$(0,0,0)$
222	$[3222]_4 = 234$	$(0,0,0)$
230	$[3230]_4 = 236$	$(0,1,0)$
301	$[3301]_4 = 241$	$(1,0,0)$
302	$[3302]_4 = 242$	$(1,0,0)$
310	$[3310]_4 = 244$	$(1,1,0)$
320	$[3320]_4 = 248$	$(1,1,0)$

This enumeration is also complete. If the first letter is 1, the remaining suffix of length 2 must be one of

$$11,\ 12,\ 21,\ 22,\ 30,$$

which gives

$$111,\ 112,\ 121,\ 122,\ 130.$$

The same argument for the first letter 2 gives

$$211,\ 212,\ 221,\ 222,\ 230.$$

If the first letter is 3, the first prefix has weight 1, so the remaining two letters must have total weight -1. The possibilities compatible with nonnegative prefix weights are

$$01,\ 02,\ 10,\ 20,$$

giving

$$301,\ 302,\ 310,\ 320.$$

A word beginning with 0 is impossible. Hence the table contains all

$$5 + 5 + 4 = 14$$

two-coloured Motzkin words of length 3.

Thus the corresponding elements of generation 4 of $\mathcal{P}$ are exactly

$$H = \{213, 214, 217, 218, 220, 229, 230, 233, 234, 236, \\ 241, 242, 244, 248\} \subseteq \mathcal{P}. \tag{21}$$

In fact, H is not merely a subset of $\mathcal{P}$: it is exactly the whole of generation 4.

Finally, Corollary 8 gives

$$\begin{aligned} g_4 &= 3 \cdot 4^3 + \frac{4^3 - 1}{3} \\ &= 192 + 21 \\ &= 213. \end{aligned}$$

Consequently, every element of $\mathcal{P}$ below 213 belongs to one of generations 1, 2, and 3, whose elements are exactly the eight members of L listed in the first table. Thus the stronger identity

$$\mathcal{P} \cap [0, 212] = L$$

holds, and in particular

$$\mathcal{P} \cap [0, 211] = L. \tag{22}$$

Lemma 15 (Finite certificate for the full family). *The following inclusions hold:*

sumset	*contained interval*
$5L$	$[215, 254]$
$H + 4L$	$[245, 486]$
$2H + 3L$	$[439, 674]$
$3H + 2L$	$[645, 862]$
$4H + L$	$[855, 1046]$.

(23)

Moreover,

$$209, 210, 211 \notin F_5(L). \tag{24}$$

Proof. Equation (23) is a finite-sumset inclusion certificate, while (24) is an exhaustive computation of $F_5(L)$ in the relevant finite range. Both are verified by the same recursive finite-set addition used in Lemma 13 and are reproduced by the supplementary verification script. □

The five intervals in Lemma 15 overlap, and hence

$$[215, 1046] \subseteq 5\mathcal{P}. \tag{25}$$

Proposition 16. *Every integer $n \geq 215$ is a sum of exactly five elements of $\mathcal{P}$. Consequently,*

$$[212, \infty) \subseteq F_5(\mathcal{P}).$$

The lower endpoint is sharp: none of $209, 210, 211$ *belongs to* $F_5(\mathcal{P})$.

Proof. By Corollary 6,

$$4 \cdot \mathcal{P} + \{1, 2\} \subseteq \mathcal{P}.$$

Equation (25) contains, in particular,

$$[215, 864] \subseteq 5\mathcal{P}.$$

Since

$$864 = 4 \cdot 215 + 5 - 1,$$

Corollary 11, applied with $k = 5$ and $M = 215$, gives

$$[215, \infty) \subseteq 5\mathcal{P}.$$

Moreover,

$$\begin{aligned}
212 &= 53 + 53 + 53 + 53, \\
213 &= 53 + 53 + 53 + 54, \\
214 &= 53 + 53 + 54 + 54,
\end{aligned}$$

so $[212, \infty) \subseteq F_5(\mathcal{P})$. Finally, (22) and (24) give

$$209, 210, 211 \notin F_5(\mathcal{P}).$$

□

5 Exact additive representation results

The first exceptional value is explained without computation.

Proposition 17. *The integer* 46 *requires exactly eight positive primitive Dyck numbers: it is a sum of eight, but it is not a sum of at most seven.*

Proof. Below 48 the only positive primitive Dyck numbers are 2 and 12. Indeed, the primitive words of length two and four are 10 and 1100, whereas every primitive word of length at least six begins with 11 and therefore represents an integer at least $[110000]_2 = 48$. Thus any representation of 46 has the form

$$46 = 12a + 2b, \qquad a, b \geq 0.$$

Equivalently, $6a + b = 23$. Since $6 \cdot 4 > 23$, we have $a \leq 3$, and hence the number of summands satisfies

$$a + b = 23 - 5a \geq 8.$$

Equality is attained by

$$46 = 12 + 12 + 12 + 2 + 2 + 2 + 2 + 2.$$

□

The decomposition

$$\mathcal{N}_{\mathrm{pD}} = \{0, 2\} \cup 4 \cdot \mathcal{P}$$

converts summand-count conditions uniformly into finite-sum conditions on $\mathcal{P}$.

Lemma 18 (Mod-4 summand criterion)**.** *Let $\epsilon \in \{0, 1\}$, let $h \geq 0$, and put $N = 4m + 2\epsilon$. Then N is a sum of at most h positive primitive Dyck numbers if and only if*

$$m \in \bigcup_{\substack{0 \leq s \leq h \\ s \equiv \epsilon \pmod 2}} \left(\frac{s - \epsilon}{2} + F_{h-s}(\mathcal{P}) \right). \tag{26}$$

Proof. Let s be the number of summands equal to 2. By Lemma 3, every remaining positive summand has the form $4p_i$ with $p_i \in \mathcal{P}$. Reduction modulo 4 gives $s \equiv \epsilon \pmod 2$, and

$$4m + 2\epsilon = 2s + 4\sum_i p_i$$

is equivalent to

$$m = \frac{s - \epsilon}{2} + \sum_i p_i.$$

There are at most $h - s$ remaining summands, which proves necessity. Conversely, any representation belonging to one of the sets in (26) gives a representation of N after replacing each p_i by $4p_i$ and adjoining s copies of 2. ☐

Corollary 19. *Let $N = 4m + 2$. Then N is a sum of at most six positive primitive Dyck numbers if and only if*

$$m \in F_5(\mathcal{P}) \cup \big(1 + F_3(\mathcal{P})\big) \cup \big(2 + F_1(\mathcal{P})\big). \tag{27}$$

Proof. Apply Lemma 18 with $\epsilon = 1$ and $h = 6$, and use $s = 1, 3, 5$. ☐

Lemma 20 (Small multiples of four)**.** *Among the positive multiples of* 4 *below* 100*, the only integer not representable by at most six positive primitive Dyck numbers is* 44*. Moreover,*

$$44 = 12 + 12 + 12 + 2 + 2 + 2 + 2.$$

Proof. The positive primitive Dyck numbers below 100 are precisely

$$2, 12, 52, 56.$$

An exhaustive computation of sums of at most six elements of this finite set, truncated below 100, shows that the only missing positive multiple of 4 is 44. The displayed identity gives a seven-summand representation. This finite check is also included in the supplementary verification script. ☐

Lemma 21 (Finite exceptional-set certificate)**.** *Let L be the set defined in* (20)*. Then*

$$\begin{aligned}&[0, 211] \setminus \Big(F_5(L) \cup (1 + F_3(L)) \cup (2 + F_1(L))\Big)\\ &\qquad = \{8, 11, 24, 38, 49, 50, 51, 209, 210, 211\}.\end{aligned} \tag{28}$$

Proof. Starting from the finite set L, compute $F_1(L)$, $F_3(L)$, and $F_5(L)$ exactly by recursive finite-set addition, truncate the result to $[0, 211]$, and take the union and its complement. This gives (28); the computation is reproduced by the supplementary verification script. ☐

Proof of Theorem 1. Suppose first that $N \equiv 0 \pmod 4$. If $N \geq 100$, write $N = 4n$. Then $n \geq 25$, so Proposition 14 and (15) give a representation of N with at most six primitive Dyck summands. Lemma 20 treats the remaining positive multiples of 4 and shows that $r(44) = 7$.

Now let $N \equiv 2 \pmod 4$, say $N = 4m + 2$. If $N \geq 850$, then $m \geq 212$, so Proposition 16 gives $m \in F_5(\mathcal{P})$. Corollary 19 gives a representation of N with at most six primitive Dyck summands. Together with the preceding paragraph, this already shows that every even $N \geq 848$ has such a representation.

It remains to classify the integers $N \leq 846$ with $N \equiv 2 \pmod 4$. Here $0 \leq m \leq 211$. By (22), Lemma 21, and Corollary 19, the corresponding values $N = 4m + 2$ that are not sums of at most six primitive Dyck numbers are precisely

$$34, 46, 98, 154, 198, 202, 206, 838, 842, 846.$$

Thus these ten numbers, together with 44, are exactly the positive even integers that are not sums of at most six primitive Dyck numbers.

All of them except 46 have seven-summand representations:

$$\begin{aligned}
34 &= 12 + 12 + 2 + 2 + 2 + 2 + 2,\\
44 &= 12 + 12 + 12 + 2 + 2 + 2 + 2,\\
98 &= 56 + 12 + 12 + 12 + 2 + 2 + 2,\\
154 &= 52 + 52 + 12 + 12 + 12 + 12 + 2,\\
198 &= 56 + 52 + 52 + 12 + 12 + 12 + 2,\\
202 &= 56 + 56 + 52 + 12 + 12 + 12 + 2,\\
206 &= 56 + 56 + 56 + 12 + 12 + 12 + 2,\\
838 &= 240 + 240 + 240 + 52 + 52 + 12 + 2,\\
842 &= 240 + 240 + 240 + 56 + 52 + 12 + 2,\\
846 &= 240 + 240 + 240 + 56 + 56 + 12 + 2.
\end{aligned}$$

Proposition 17 shows that 46 requires exactly eight summands. This proves all assertions of the theorem. □

Corollary 22. *The integer* 848 *is the least even integer T such that every even integer $N \geq T$ is a sum of at most six primitive Dyck numbers.*

Proof. Theorem 1 shows that six summands suffice for every even integer at least 848, whereas 846 requires seven summands. □

Corollary 23. *Every nonnegative even integer is a sum of at most eight elements of $\mathcal{N}_{\mathrm{pD}}$, and* 46 *is the unique even integer that is not a sum of at most seven elements of $\mathcal{N}_{\mathrm{pD}}$.*

Proof. The assertion for positive integers follows immediately from Theorem 1; the integer 0 is the empty sum. □

Following standard additive-basis terminology [7], for $A \subseteq 2\mathbb{N}_0$ we call A a *relative additive basis of order at most h for* $2\mathbb{N}_0$ if every element of $2\mathbb{N}_0$ is a sum of at most h elements of A. Its relative order is *exactly* h if h is the least integer with this property. We similarly call A a *relative asymptotic additive basis of order at most h* if every sufficiently large element of $2\mathbb{N}_0$ is a sum of at most h elements of A.

Theorem 1 gives the complete exceptional set for six-summand representability. Corollary 23 shows that $\mathcal{N}_{\mathrm{pD}}$ is a relative additive basis of exact order eight for $2\mathbb{N}_0$, while Corollary 22 identifies the sharp six-summand threshold. The next section explains the difference between the two residue classes and proves that the relative asymptotic order is exactly six.

6 Generation gaps and asymptotic optimality

For multiples of 4, the five-summand representations in $\mathcal{P}$ may be used directly. For integers congruent to 2 modulo 4, however, an odd number of copies of the exceptional summand 2 must be used. This distinction raises the relative asymptotic order of the full primitive family from 5 to 6.

Corollary 24 (Five-summand criterion)**.** *Let $N = 4m + 2$ with $m \geq 3$. Then $r(N) \leq 5$ if and only if*

$$m \in F_4(\mathcal{P}) \cup \big(1 + F_2(\mathcal{P})\big). \tag{29}$$

Proof. Apply Lemma 18 with $\epsilon = 1$ and $h = 5$. The values $s = 1, 3, 5$ give

$$F_4(\mathcal{P}) \cup \big(1 + F_2(\mathcal{P})\big) \cup \{2\}.$$

The last set is irrelevant because $m \geq 3$. □

The following lemma isolates the lower-bound mechanism created by the gaps between generations.

Lemma 25 (Recurring obstruction)**.** *For every integer $k \geq 2$, put*

$$x_k = 10 \cdot 4^k - 2.$$

Then

$$x_k \notin F_4(\mathcal{P}) \qquad \textit{and} \qquad x_k - 1 \notin F_2(\mathcal{P}). \tag{30}$$

Proof. Fix $k \geq 2$, and abbreviate $x = x_k$. The least element of generation $k+2$ is

$$g_{k+2} > 3 \cdot 4^{k+1} = 12 \cdot 4^k > x.$$

Consequently, every element of $\mathcal{P}$ that is at most x has generation at most $k+1$. By Corollary 8, every such element is either *small*, of generation at most k and hence at most

$$U_k = 4^k - 2^{k-1},$$

or *large*, of generation $k+1$ and hence between g_{k+1} and $U_{k+1} = 4^{k+1} - 2^k$.

First consider $x - 1 = 10 \cdot 4^k - 3$. It is larger than U_{k+1} and smaller than g_{k+2}, so it does not itself belong to $\mathcal{P}$. Moreover, any sum of two eligible elements is at most

$$2U_{k+1} = 8 \cdot 4^k - 2^{k+1} < 10 \cdot 4^k - 3 = x - 1.$$

Thus $x - 1 \notin F_2(\mathcal{P})$.

Now suppose, for contradiction, that

$$x = q_1 + \cdots + q_t, \qquad t \leq 4, \qquad q_i \in \mathcal{P}.$$

Let N_{b} be the number of large summands. If $N_{\mathrm{b}} \leq 2$, then filling the remaining positions with the largest possible small summands gives

$$\begin{aligned} x &\leq 2U_{k+1} + 2U_k \\ &= 2(4^{k+1} - 2^k) + 2(4^k - 2^{k-1}) \\ &= 10 \cdot 4^k - 3 \cdot 2^k < 10 \cdot 4^k - 2 = x, \end{aligned}$$

a contradiction. Hence $N_{\mathrm{b}} \geq 3$. But then the three large summands alone have sum at least

$$3g_{k+1} = 10 \cdot 4^k - 1 = x + 1$$

by (9), again a contradiction. Therefore $x \notin F_4(\mathcal{P})$. □

Corollary 26 (Order of $\mathcal{P}$). *The set $\mathcal{P}$ is an asymptotic additive basis of order exactly 5 for the nonnegative integers. More precisely, every integer $n \geq 212$ lies in $F_5(\mathcal{P})$, whereas $F_4(\mathcal{P})$ and $F_3(\mathcal{P})$ each omit infinitely many integers.*

Proof. Proposition 16 gives $[212, \infty) \subseteq F_5(\mathcal{P})$. For every $k \geq 2$, Lemma 25 gives $x_k \notin F_4(\mathcal{P})$. Since $x_k \to \infty$, four summands do not suffice asymptotically. The same is true for three summands because $F_3(\mathcal{P}) \subseteq F_4(\mathcal{P})$. □

Corollary 27 (Multiples of 4 requiring five summands). *For every integer $k \geq 3$,*

$$M_k = 10 \cdot 4^{k+1} - 8$$

satisfies $r(M_k) = 5$. Hence infinitely many multiples of 4 require exactly five positive primitive Dyck summands.

Proof. Write $M_k = 4x_k$. Since $k \geq 3$, one has $x_k \geq 212$, and Proposition 16 gives $x_k \in F_5(\mathcal{P})$. Multiplying such a representation by 4 shows that $r(M_k) \leq 5$.

Suppose that $r(M_k) \leq 4$. Applying Lemma 18 with $\epsilon = 0$ and $h = 4$ gives

$$x_k \in F_4(\mathcal{P}) \cup \big(1 + F_2(\mathcal{P})\big) \cup \{2\}.$$

The first two alternatives contradict Lemma 25, and the last is impossible because $x_k > 2$. Thus $r(M_k) \geq 5$, and equality follows. □

Proof of Theorem 2. Fix $k \geq 2$ and put $m = x_k = 10 \cdot 4^k - 2$. Then

$$N_k = 4m + 2 = 10 \cdot 4^{k+1} - 6.$$

By Lemma 25,

$$m \notin F_4(\mathcal{P}), \qquad m - 1 \notin F_2(\mathcal{P}).$$

Corollary 24 therefore gives $r(N_k) \geq 6$. The integer N_k is not one of the eleven exceptions in Theorem 1, so $r(N_k) \leq 6$. Hence $r(N_k) = 6$. Since $N_k \to \infty$, no $h \leq 5$ bounds $r(N)$ for all sufficiently large even N, while Theorem 1 gives the upper bound $h = 6$. Therefore $h_{\text{asym}} = 6$. □

Remark 28 (The two residue classes). If $N = 4n$, then $n \in F_5(\mathcal{P})$ for all sufficiently large n, so $r(N) \leq 5$ eventually; Corollary 27 shows that equality occurs infinitely often. If $N = 4m + 2$, the number of copies of 2 must be odd. For $m = x_k$, the one-copy case would require $m \in F_4(\mathcal{P})$ and the three-copy case would require $m - 1 \in F_2(\mathcal{P})$; Lemma 25 excludes both. This residue class raises the relative asymptotic order from 5 to 6.

7 Effective verification and reproducibility

The infinite assertions in this paper are proved symbolically. Computation is used only for explicitly displayed finite certificates and for an independent check of the representation function.

For a finite set $X \subseteq \mathbb{Z}_{\geq 0}$, a summand count k, and a cutoff U, exact membership in $jX \cap [0, U]$ is computed recursively by

$$S_0 = \{0\}, \qquad S_{j+1} = (S_j + X) \cap [0, U] \quad (0 \leq j < k).$$

With Boolean arrays, this requires $O(kU|X|)$ time and $O(U)$ working space. Mixed certificates such as $2H + 3L$ are checked by the same recursion with the prescribed finite set at each addition step.

Every contained interval is then scanned exhaustively, and every claimed complement is computed exactly.

The two Python source files used for the finite checks are available at the following commit-pinned links and can also be distributed with the manuscript as supplementary material. The program `primitive_dyck_representation.py` at commit `942c264` enumerates primitive Dyck numbers and computes $r(N)$ by unbounded coin-change dynamic programming. Its default run generates the first 50,000 values $r(2n)$ and checks all indices corresponding to the values seven and eight in Theorem 1.

The finite certificates are reproduced by `verify_certificates_for_paper.py` at commit `942c264`. The program uses only exact integer arithmetic and the Python standard library. It verifies (17)–(18), (23), (24), the small multiples of four, and the finite exceptional-set identity (28). It also performs an independent dynamic-programming cross-check of $r(N)$ up to a user-selected bound.

In the recorded reference run under CPython 3.14.6, the programs generated the first 50,000 values of the representation function and checked all eleven indices with value at least seven. They also reproduced every finite certificate, independently verified $r(N)$ through $N = 50{,}000$, and checked the recurring obstruction and exact-order instances through $k = 5$.

8 Conclusion

Primitive Dyck words provide a natural boundary example for combining regular approximation with genuinely context-free structure. In the proof developed here, the regular language 11(01 | 10)*00 supplies the covering for one residue class, while the exact positional Motzkin coding of the full deterministic context-free language supplies the covering for the other. The interval digit-lifting theorem turns finite arithmetic certificates into infinite tails and, constructively, into logarithmic-depth representation algorithms. The gaps between Motzkin-coded generations yield recurring lower-bound obstructions.

The resulting value set has three exact additive orders. Its relative order on the nonnegative even integers is eight, its relative asymptotic order is six, and the halved family $\mathcal{P}$ has asymptotic additive order five. The sharp six-summand threshold is 848, but six summands remain necessary infinitely often.

The method suggests two broader directions. First, one may seek other nonregular language-defined sets admitting an arithmetically effective regular underapproximation together with a complementary block coding of the full language. Second, the interval digit-lifting principle can be studied for other bases and digit intervals, with the finite seed certificates synthesized automatically. For the present family, a more specific open problem is to characterize all even N satisfying $r(N) = 6$; computations indicate further block structure near $10\cdot 4^k$, but its exact boundaries remain unknown.

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